Yi Zhang(Joey)

Tenured Associate Professor · Xiamen University

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EDUCATION

National Taiwan University
Ph.D. in Communication Engineering — *Sept. 2017 - Jul. 2020*

National Taiwan University
M.S. in Communication Engineering — *Sept. 2014 - Jul. 2016*

Xiamen University
B.S. in Software Engineering — *Sept. 2010 - Jun. 2014*

WORK EXPERIENCE

School of Informatics, Xiamen University — **Xiamen, China**
Tenured Associate professor with Dept. of Info. and Comm. Engineering — *Aug. 2025 - Current*

School of Informatics, Xiamen University — **Xiamen, China**
Assistant professor with Dept. of Info. and Comm. Engineering — *Nov. 2020 - Jul. 2025*

Haixi Institutes, Chinese Academy of Sciences — **Quanzhou, China**
Assistant engineer with Quanzhou Institute of Equipment Manufacturing — *Nov. 2016 - May 2017*

Chinasoft International Limited — **Xiamen, China**
Student internship — *Jul. 2013 – Aug. 2013*

Xiamen University — **Xiamen, China**
Team member of RCS robot team — *Aug. 2012 – Jul. 2013*

HONORS & AWARDS

- **2025** Research on “Air-Ground Cooperative Communication-Navigation Integration” is fund by Youth Science Foundation of National Natural Science Foundation of China (NSFC)
- **2022** Research on “End-Edge-Cloud Dynamic Resource Allocation in Cellular Vehicular Networks” is funded by the Natural Science Foundation of Fujian Province of China
- **2021-2024** Jointly-appointed postgraduate supervisor in Institute of Artificial Intelligence, Xiamen University
- **2023** C-class high-level personnel, Fujian Province
- **2022** C-class high-level personnel, Xiamen City
- **2021** High-level overseas-educated personnel, Xiamen City
- **2017** Core member of high-level talent team (specialized in "IoT Technology Based on Optical Fiber Sensing") secured Quanzhou’s Talent "Harbor Plan"
- **2016** Fifth class outstanding person award of Jinjiang City
- **2013** Embeded system designer of the national technical qualification examination for professional computer and software technologies, China
- **2013** The second award of domestic contest of ABU Asia-Pacific Robo Contest (ABU Robocon), China

SERVICE & LEADERSHIP

- Senior Member of China Institute of Communications (CIC)
- Professional Member of IEEE
- Vice President (2022-2024) and AC (2021-2024) of Xiamen Chapter of CCF YOCSEF
- TPC of ICA3PP 2021 (CCF Rank-C conference), BlockchainEvo 2022, BigCom 2022, ICICA 2022, BigCom2023, ICICA 2023, WCNC 2024, ICICA 2024, WCNC 2025, AIoT 2025
- Wireless AI and Communication-Sensing Integration task groups of the IMT-2030 (6G) Promotion Group
- Organizing Committee of IEEE/CIC ICCC 2021
- *Journal article reviewer* IEEE Transactions on Mobile Computing (TMC), IEEE/ACM Transactions on Networking(TNET), IEEE Transactions on Dependable and Secure Computing (TDSC), IEEE Transactions on Wireless Communications (TWC), IEEE Transactions on Communications (TCOM), IEEE Internet of Things Journal (IoT-J), IEEE Transactions on Vehicular Technology (TVT), IEEE Transactions on Network Science and Engineering (TNSE), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Industrial Informatics (TII), IEEE Systems Journal (ISJ), IEEE Communications Letters (COMML), China Communications (CNCOMM), IEEE Access, IEEE/CAA Journal of Automatica Sinica (JAS), Wireless Communications and Mobile Computing (WCMC)
- *Conference & proceeding paper reviewer* IEEE Vehicular Technology Conference (VTC), IEEE Vehicular Networking Conference (VNC), International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP), International Conference on Big Data Computing and Communications (BigCom), Conference on Web and Internet Economics (WINE), International Conference on Information Communication and Applications (ICICA), IEEE Annual Congress on Artificial Intelligence of Things (AIoT)

TEACHING

- **Fall 2022, Fall 2023, Fall 2024, Fall 2025** Embedded Microcontroller *Engineering Elective*
- **Spring 2021, Spring 2022, Spring 2023, Spring 2024, Spring 2025** Foundations of Network Security: Attack and Defense *Major Core* Joint program with ANT SECURITY LAB of ANT GROUP and QI-ANXIN
- **Fall 2021, Fall 2022, Fall 2023** Foundamentals of Information Security *Disciplinary Elective* Cooperative teaching with Prof. Liang Xiao
- **Fall 2021** Wireless Communication Networks *Postgraduate* Cooperative teaching with Prof. Liang Xiao

RESEARCH INTERESTS

- Mobile and wireless networking
- Fog/edge computing and networking
- Game theory models, pricing, and incentive mechanism design

SELECTED PUBLICATIONS

Books

[1] L. Xiao, **Y. Zhang**, X. Lu, and H. Yang, *Wireless Communication Security: Principles and Practice*. China: Tsinghua University Press, 2025.

Journals

[1] **Y. Zhang***, Y. Xie, L. Wang, M. Liwang, and X. Wang, "Reconfigurable intelligent surface enhanced wireless localization: Phase optimization for malicious interference mitigation," *IEEE Transactions on Communications*, vol. 73, no. 10, pp. 8824–8839, 2025.

[2] Y. Xiao, L. Xiao, K. Wan, H. Yang, **Y. Zhang***, Y. Wu, and Y. Zhang, "Reinforcement learning based energy-efficient collaborative inference for mobile edge computing," *IEEE Transactions on Communications*, vol. 71, no. 2, pp. 864–876, 2023.

[3] L. Xiao, H. Li, S. Yu, **Y. Zhang***, L.-C. Wang, and S. Ma, "Reinforcement learning based network coding for drone-aided secure wireless communications," *IEEE Transactions on Communications*, vol. 70, no. 9, pp. 5975–5988, 2022.

[4] **Y. Zhang** and H.-Y. Wei, "Risk-aware cloud-edge computing framework for delay-sensitive industrial iots," *IEEE Transactions on Network and Service Management*, vol. 18, no. 3, pp. 2659–2671, 2021.

[5] **Y. Zhang**, J. H. Liu, C. Y. Wang, and H. Y. Wei, "Decomposable intelligence on cloud-edge iot framework for live video analytics," *IEEE Internet of Things Journal*, vol. 7, no. 9, pp. 8860–8873, 2020.

[6] **Y. Zhang**, Z. Chen, and H. I. Hee, "Noninvasive Measurement of Heart Rate and Respiratory Rate for Perioperative Infants," *Journal of Lightwave Technology*, vol. 37, no. 11, pp. 2807–2814, June 2019.

[7] **Y. Zhang**, C. Wang, and H. Wei, "Parking Reservation Auction for Parked Vehicle Assistance in Vehicular Fog Computing," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 4, pp. 3126–3139, April 2019.

[8] **Y. Zhang**, C. Wang, and H. Wei, "Incentive Compatible Overlay D2D System: A Group-Based Framework without CQI Feedback," *IEEE Transactions on Mobile Computing*, vol. 17, no. 9, pp. 2069–2086, Sep. 2018.

[9] Y. Chiang, **Y. Zhang**, H. Luo, T.-Y. Chen, G.-H. Chen, H.-T. Chen, Y.-J. Wang, H.-Y. Wei, and C.-T. Chou, "Management and orchestration of edge computing for iot: A comprehensive survey," *IEEE Internet of Things Journal*, vol. 10, no. 16, pp. 14 307–14 331, 2023.

Conferences

[1] H. Yu, G. Huang, **Y. Zhang***, W. Cao, and L. Chen, "Terrain-driven uplink localization in mountainous environments based on moving uav anchors," in *2026 IEEE International Conference on Communications* (*ICC*), 2026, pp. 1–6.

[2] C. Zhang, G. Zheng, W. Cao, and **Y. Zhang***, "Joint optimization of role switching and trajectory planning for multi-uav assisted uplink secure communications," in *2026 IEEE International Conference on Communications Workshops* (*ICC Wkshps*), 2026, pp. 1–6.

[3] J. Li, **Y. Zhang***, Y. Chen, and H. Yu, "Cl2f: Close-loop communication and localization framework for high-mobility uav swarm networks in partial gnss-denied environments," in *2025 IEEE Annual Congress on Artificial Intelligence of Things* (*AIoT*), 2025, pp. 799–803.

[4] H. Yu, L. Wang, **Y. Zhang***, and Y. Chen, "Low-latency relay selection for ocean monitoring in buoy-based maritime wireless networks," in *2025 IEEE 102nd Vehicular Technology Conference* (*VTC2025-Fall*), 2025, pp. 1–6.

[5] H. Hong, **Y. Zhang***, and Y. Xie, "Energy-limited uav visiting planning for age-aware wireless-powered sensor networks," in *2023 IEEE 98th Vehicular Technology Conference* (*VTC2023-Fall*), 2023, pp. 1–6.

[6] **Y. Zhang**, C.-Y. Wang, and H.-Y. Wei, "Parked vehicle assisted vfc system with smart parking: An auction approach," in *2018 IEEE Global Communications Conference* (*GLOBECOM*), 2018, pp. 1–7.

[7] **Y. Zhang**, C.-Y. Wang, and H.-Y. Wei, "Incentive compatible mode selection and spectrum partitioning in overlay d2d-enabled network," in *2015 IEEE Globecom Workshops* (*GC Wkshps*), 2015, pp. 1–6.